

Total No. of Questions : 8]

SEAT No. :

PE2739

[Total No. of Pages : 2

[6583]-320

T.E. (Computer Engineering) (Honors)

VIRTUAL REALITY AND AUGMENTED REALITY

Augmented Reality

(2019 Pattern) (Semester - VI) (310703)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) What are the major software components of Augmented reality systems?[9]

b) Explain simultaneous localization and mapping in details. [8]

OR

Q2) a) Describe the various software components used in AR. [9]

b) Explain natural feature tracking by detection in AR. [8]

Q3) a) What are different types of Markers in AR? Explain the Template markers.[6]

b) Explain the term mathematical representation of matrix multiplication.[6]

c) Write a short note on Location based augmentation and visual tracking.[6]

OR

Q4) a) When to use marker based tracking? Explain. [6]

b) Write a note on [6]

i) 2D barcode markers

ii) Hybrid tracking.

c) Explain the term feature based tracking. [6]

P.T.O.

Q5) a) What are different projection displays? **[8]**

b) Write a short note on **[9]**

i) Scene generation

ii) Optical see-through HMD

iii) Virtual Retina System

OR

Q6) a) Describe the various AR components in detail. **[8]**

b) Enlist the AR devices and explain Video See through systems. **[9]**

Q7) a) Explain the term computer vision and mixed reality. **[10]**

b) Write a short note on Dense tracking and mapping(DTAM). **[8]**

OR

Q8) a) Describe the term semi-direct monocular visual odometry(SVO). **[10]**

b) Explain the term PTAM in detail. **[8]**

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